RAG -- Retrieval-Augmented Generation

Why RAG?

LLM Drawback

1. LLM cut off Date → Knowledge cut off Date ✓
2. LLM's were trained with only public avalible data ✓

Chatbot → Answer customer @uestions

Private Data X

company specific Data X

What is RAG:

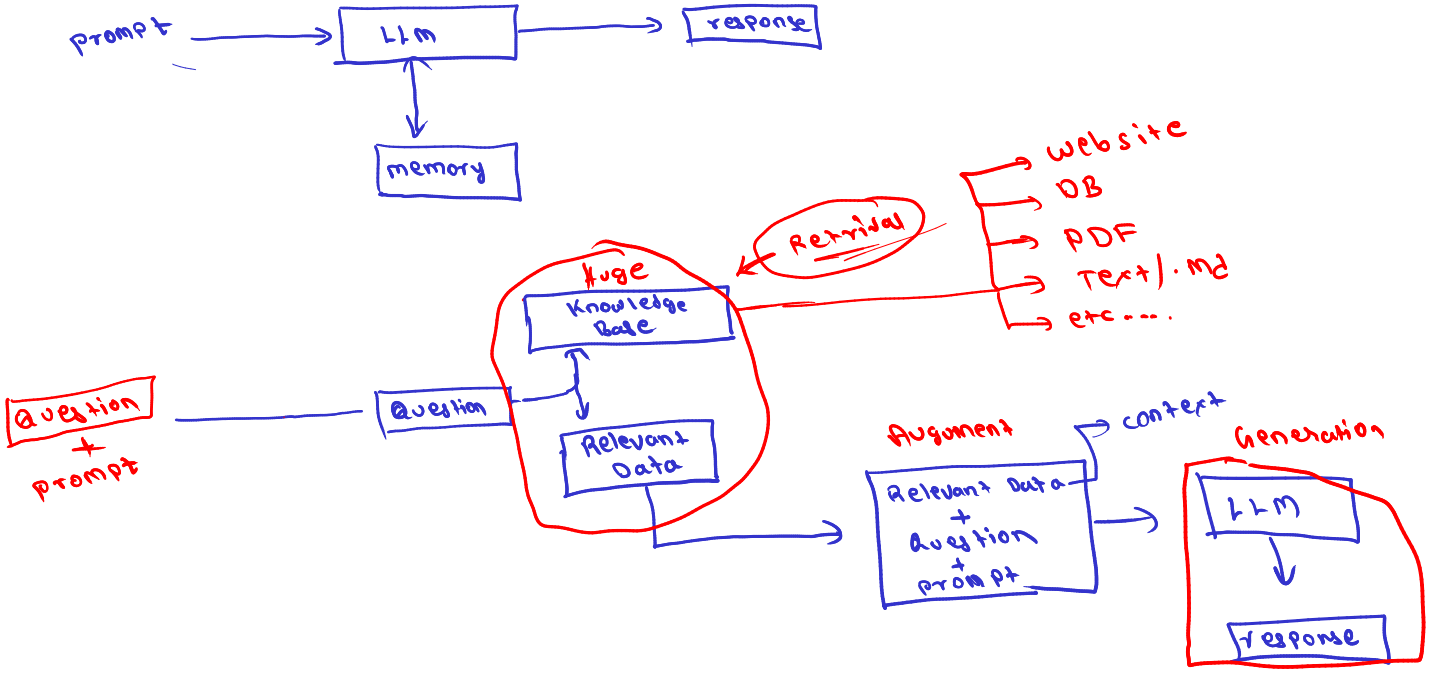
Retrieval-Augmented Generation (RAG) is a technique that gives Large Language Models (LLMs) access to external, private, or real-time data.
It prevents AI from hallucinating and ensures accuracy.

Retrieval--
Find relevant information from your specific data sources based on the question.

Augmented--
Add that relevant info to the user's prompt as context.

Generation--
The LLM generates a precise answer based on that context.

How



1. LLM has token limit -

2. cost → pay per token - if token increase then the cost increase (closed-sourced)

3. computation increase - (open-sourced)

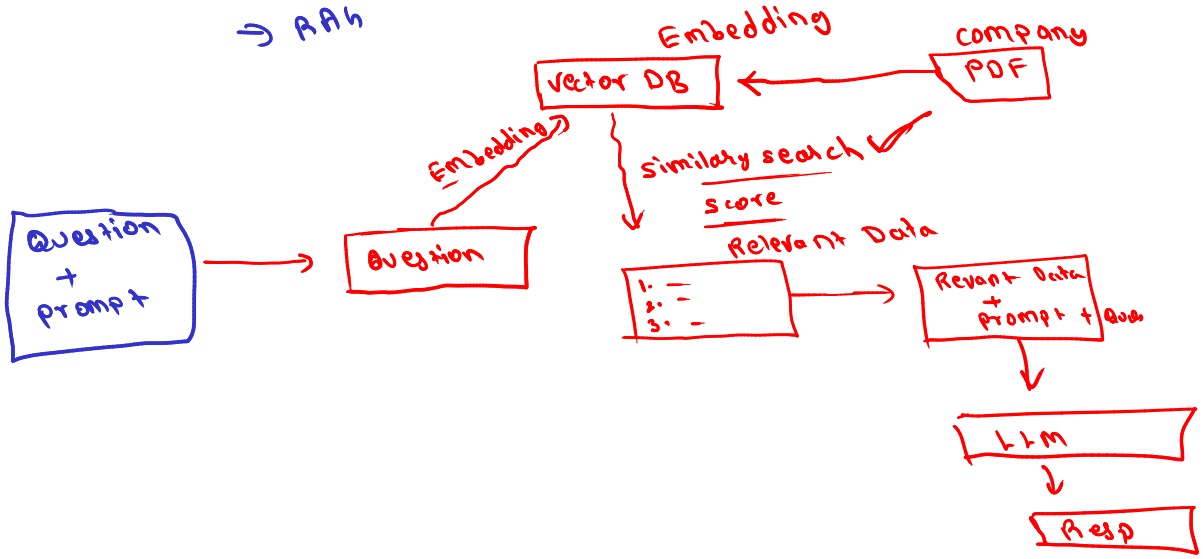
4. LLM sometime are not that good when you send large irrevant Data

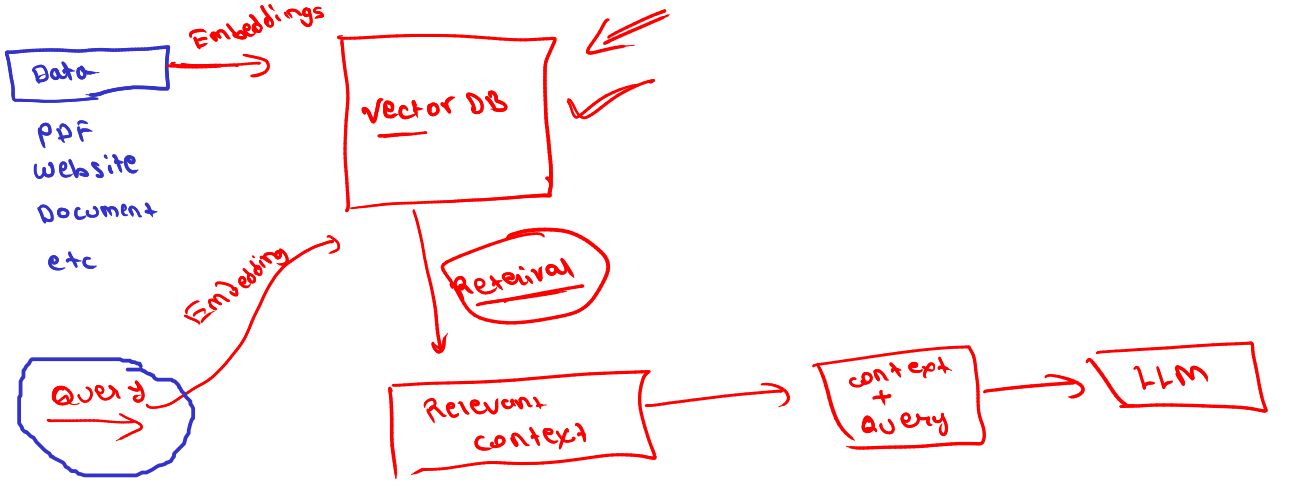
→ not a good approch to send entire Data to the LLM - for answering a specific question

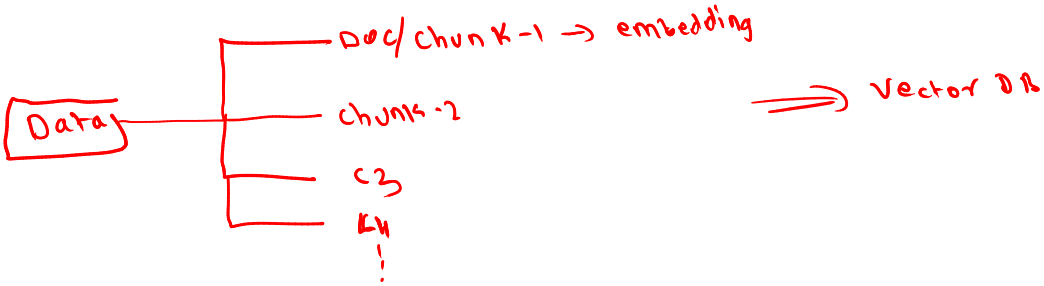
→ Similarity search

→ Vector
→ Embedding
→ cosine similarity
→ vector DB
→ RAG

RAG components







Why Text Splitting Matters:

* LLM Context Limits: Most LLMs have token limits
* Semantic Preservation: Good splitting maintains meaning and context
* Retrieval Quality: Proper chunks improve vector search accuracy
* Processing Efficiency: Smaller chunks are easier to process and embed

CharacterTextSplitter -

1. Split text using ONE separator
2. Iterate through the resulting splits
3. Greedily merge splits into chunks until chunk_size would be exceeded
4. If adding a split exceeds chunk_size:
   - finalize the current chunk
   - start a new chunk with the split
5. If a single split itself exceeds chunk_size:
   - keep it intact (chunk may exceed limit)
6. After all chunks are formed, apply chunk_overlap between consecutive chunks

Use Cases:

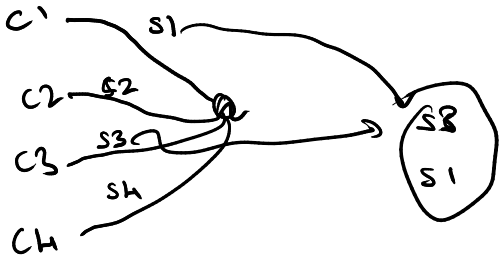
1. Simple Documents: Plain text with clear paragraph breaks
2. Quick Prototyping: Fast setup for basic text processing
3. Uniform Content: Documents with consistent structure

RecursiveCharacterTextSplitter

1. Start with the first separator in the list
2. Split text at all occurrences of this separator
3. For each resulting piece:
   - If it's within chunk_size, keep it
   - If it's too large, try the next separator recursively
4. Continue until all pieces are within chunk_size or no separators remain
5. Add overlap between consecutive chunks

Use Cases:

1. Structured Documents: Markdown, technical docs, articles
2. Multi-level Content: Documents with headers, sections, subsections
3. Semantic Preservation: When maintaining context is crucial



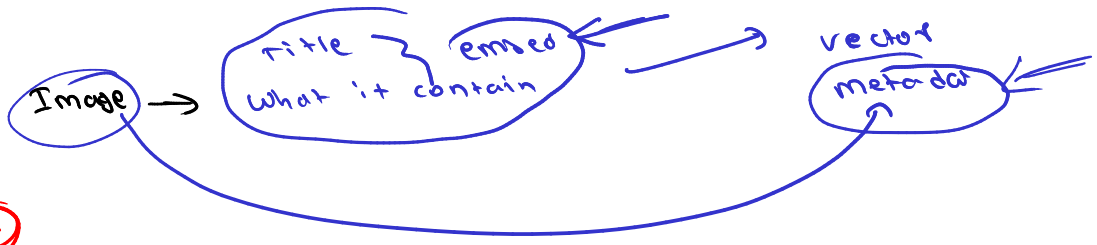
Artificial Intelligence (AI) has revolutionized the way we interact with technology.
Machine learning, a subset of AI, enables computers to learn from data without explicit programming.

Deep learning uses neural networks with multiple layers to process complex patterns.
Natural language processing (NLP) helps computers understand and generate human language.

Computer vision enables machines to interpret visual information from the world.
These technologies work together to create intelligent systems that can perform tasks previously thought to be exclusive to humans.

Robotics combines AI with mechanical engineering to create autonomous machines.
These robots can navigate environments, manipulate objects, and make decisions based on sensor data.

The future of AI holds promise for solving complex problems in healthcare, climate change, and scientific research. However, ethical considerations must be addressed to ensure responsible development and deployment of AI systems.



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Cats are small domesticated animals. They are often kept as pets.
They like to chase mice.

The stock market fluctuates daily. Investors analyze trends.
Economic factors influence prices.

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Sentence	Embedding
S1	[0.9, 0.8, 0.7]
S2	[0.88, 0.79, 0.69]
S3	[0.87, 0.78, 0.68]
S4	[0.2, 0.3, 0.4]
S5	[0.21, 0.31, 0.39]
S6	[0.22, 0.29, 0.38]

S1 → S2	0.99
S2 → S3	0.99
S3 → S4	0.45
S4 → S5	0.98
S5 → S6	0.97

Chunk 1:
Cats are small domesticated animals.
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Chunk 2:
The stock market fluctuates daily.
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Customer service Chatbot - Solar Nova

